

Milestone 00 — Project Structure Refactor

Establishing a Reproducible Framework for Regional Skew Estimation

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1 Overview of v0.4

This document describes the structure and metadata setup completed for **Milestone v0.4**.

This document tracks the **project structure refactor** for a project to estimate regional skew of the annual flood series for the Great Plains ecoregion. The reorganization improves clarity, modularity, and reproducibility by aligning files with milestone workflows and standard naming conventions.

Additionally, this document tracks the **development of a structured covariate metadata file** to support reproducibility, clarity, and consistency in modeling regional skew. The metadata defines each covariate’s name, description, units, analytical resolution, and conceptual grouping.

1.1 Covariate Metadata

1.1.1 Purpose

The covariate metadata documents all variables used in regional skew modeling and spatial stratification. It is designed to ensure:

- consistent naming across scripts and visualizations,
- interpretable plots and maps,
- scalable use across spatial resolutions and modeling domains.

1.1.2 Source

The original .xlsx file is stored as: docs/skew_covariates_metadata_v01.xlsx

Older working versions are archived in: docs/delete_later/

1.1.3 Script and Output Location

The script used to generate modular .csv files: docs/01_split-xlsx-into-csv.R

Output .csv files (one per worksheet): docs/covariates_metadata_split/

1.1.4 QA / Validation

A multi-step metadata quality checklist was used to validate content before export. This includes checks for:

- completeness of key fields,
- consistency in phrasing and formatting,
- conceptual alignment across scales and groups,
- clarity and interpretability of variable description

2 Summary of Changes

This milestone documents a major restructuring of the project folder hierarchy, archiving legacy files, and organizing scripts into modular milestone-based subfolders. Additionally, this milestone introduces the hierarchical covariate framework and macrozone classification for Great Plains analysis and develops covariate metadata. The following changes were made to the project:

- Soft-archived old scripts and notebooks
- Modularized R/ folder structure with numbered milestones
- Consolidated utility functions into a reusable location
- Moved and modularized R scripts into milestone-based folders
- Moved and renamed and organized shapefiles for clarity
- Reorganized this Rmd to improved clarity
- Identified and documented covariates with metadata
- Replaced legacy .gitignore with spatial-aware template
- Standardized naming to improve long-term maintainability
- Developed a consistent naming strategy for data folders and files
- Refactored this milestone – moved download-related steps to milestone_01

3 Project structure (v0.3)

3.1 Original project structure

```
FFA_regional-skew/  
  arcgis_project/    # ArcGIS files  
  data/  
    raw/             # Raw downloads (PRISM, USGS, etc.)  
    clean/           # Cleaned covariate & skew datasets  
    meta/            # # Metadata for reproducibility  
    spatial/         # Spatial data for ecoregions (EPA)  
  figures/           # Plots and heatmaps  
  functions/         # Reusable R functions  
  references_pdfs/    # Background readings
```

```

model_summaries/ # CSVs of model coefficients, VIFs, etc.
results/         # Rmd files from prior EDA
scripts/         # Modular R scripts by milestone
README.Rmd       # Workflow overview (editable)
README.md        # Rendered Markdown output
.gitignore       # Prevents sensitive/local files from being pushed

```

3.2 Updated File Structure (v0.3)

```

FFA_regional-skew/
.gitignore           # Prevents sensitive/local files from being pushed
arccgis_project/     # Stores `.aprx` and layer files from ArcGIS
                    #   Pro workflows

data/
  raw/               # Unmodified input data
                    #   e.g., shapefiles, CSVs, rasters
    elevation_ned.tif
    sites_all_in_bb.csv
    sites_all_peak_in_bb.csv
    raster_raw/      # PRISM rasters
    vector_raw/      # Shapefiles (unprojected/unprocessed)
      ecoregions_orig/
      ecoregions_unprojected/
      koppen-climate-classification/
      us_eco_lev01/ through us_eco_lev04/
      misc shapefiles/

  processed/         # Cleaned, derived datasets

    spatial/         # Ready-to-use shapefiles and rasters
      us_eco_lev01/
      us_eco_lev02/
      us_eco_lev03/
      us_eco_lev04/
      koppen-climate/
      tl_state_boundaries/
      derived_products/

  meta/              # CRS info, variable scaffold, project metadata
    variable_scaffold.csv

docs/                # Project documentation, e.g., final reports,
                    #   manuscripts, proposal materials.
                    # Reference documentation like README-style guides.
                    # Metadata crosswalks and data dictionaries
                    # review or publication.
                    # Files you reference in Quarto/PDF reports or posters

FFA_regional-skew.Rproj # RStudio project file for launching the
                       # workspace. Keep this in the root.

log/                 # For shell logs or targets progress reports
notebooks/           # For ad hoc .Rmd or .qmd experiments
notes/               # Personal or team notes, meeting logs, brainstorm

```

```

# Could be transitioned to Markdown or Quarto as
# the project matures

output/
# Intermediate outputs (e.g., `.Rds`, `.csv`, `.tif`)
# Next Steps: Add subfolders like `extracted/`,
#   `joined/`, or date-stamped folders |

  figs/
  models/
  tables/
# Plots and maps
# Model objects (.rds)
# Summary tables (.csv, .html)

R/
# All analysis scripts (milestone-organized)
  01_download/
  02_clean/
  03_covariates/
  04_modeling/
  05_eval/
  utils/
  f_process_geometries.R
# NWIS, PRISM, Ecoregions
# Filtering, QA, station skew
# Climate, topography, land cover
# GAMs, Elastic Net, correlation
# Model diagnostics, residuals, validation
# Reusable functions

README.md
# Rendered Markdown output. GitHub-compatible
# plain-text overview. Use for quick
# navigation, build instructions, etc.

README.Rmd
# Workflow overview (editable). Richer,
# knit-ready documentation with figures,
# tables, and references. Can generate
# HTML/PDF documentation from this file

reports/
# analysis narratives, usually knitted `.Rmd`
# or `.qmd` output. Next Steps: Consider
# `reports/final/`, `reports/draft/`
# structure if versioning

results/
# Manuscript-ready outputs, model metrics,
# final figures, tables, model outputs for
# publication or reporting Next Steps:
# Organize by milestone or product:
#   `maps/`, `tables/`, `models/`

  posterdown/
  slides/
# Poster files and assets
# Slide decks or visualizations

to_check/
# Temporary holding area for uncertain or
# transitional files needing review or QA.
# Next Steps: Consider renaming to `sandbox/`
# and clearing regularly.

```

4 Refactor project structure (v0.4)

4.1 Migration Checklist – v0.4

Step	Task	Status
0.3.1	Soft archive old files	[X]
0.3.2	Create folders for each milestone step	[X]
0.3.3	Move functions/ to R/utils/ and source with source()	[X]
0.3.4	Rename scripts/ to R/ and split by milestone function	[X]
0.3.5	Move raw shapefiles to data/raw/spatial_raw/	[X]
0.3.6	Move cleaned shapefiles to spatial/	[X]
* 0.4 *	Identified a need to update covariates	[X]
0.4.1	Reorganize this Rmd	[X]
0.4.2	Identify and document covariates	[X]
0.4.3	Validate and document Notes folder	[X]
0.4.4	Update raw data/ folder structure / naming strategy	[X]
0.3.7	** Updated ** Move steps to milestone_01	[X]
0.3.8	Update GitHub Milestone and Issue	[X]
0.3.9	Document final changes in this .Rmd	[X]
0.3.10	Document final changes in README.Rmd	[X]
0.3.11	Tag this commit in Git as v0.3-structure-refactor	[X]

4.2 v0.3 & v0.4 Step-by-Step Refactor Log

4.2.1 Step 0.3.1 - Soft archive old files

- **Action:** Move unverified/legacy files to to_check/ or delete/ with caution.
- **Before:** Unorganized files that may or may not be aligned with current project workflow.
- **Reason:** Improves clarity, modularity, and reproducibility of workflow.
- **After:** Cleaner directory structure.

4.2.2 Step 0.3.2 — Create files for each milestone step

- **Reason:** Improves clarity, modularity, and reproducibility of workflow.
- **Before:** Files partially aligned with milestone workflows. Ad hoc naming convention.
- **After:** Separate raw/, clean/, meta/, functions/, and output/. Spatial data organized and modularized.

4.2.3 Step 0.3.3 - Move functions into a unified utils/ folder

- **Reason:** Centralize and reuse processing functions.
- **Before:** functions/f_process_geometries.R
- **After:** R/utils/f_process_geometries.R
- **Scripts affected:** Any sourcing functions/; update to source("R/utils/...")

4.2.4 Step 0.3.4 — Split and organize scripts/ into milestone-based folders

- **Reason:** Improves modularity and tracks workflow by purpose.
- **Before:** All code in scripts/
- **After:** R/utils/f_process_geometries.R

```
R/
  01_download/
  02_clean/
  03_covariates/
  04_modeling/
  05_eval/
```

- **Scripts affected:** All main workflow scripts.

4.2.5 Step 0.3.5 — Move raw spatial and climate data into data/raw/

- **Reason:** Distinguish between raw inputs and processed outputs.
- **Before:** data/spatial/ held both raw and derived data
- **After:** R/utils/f_process_geometries.R

```
data/
  raw/
    raster_raw/      # Ecoregions, Koppen, PHZM, etc.
    vector_raw/
```

- **Folders moved:** ecoregions, koppen-climate-classification, phzm, us_eco

4.2.6 Step 0.3.6 — Promote Processed Spatial Layers to /spatial/

- **Reason:** Maintain easy access to ready-to-use shapefiles.
- **Before:** Mixed with raw
- **After:** R/utils/f_process_geometries.R

```
spatial/
  us_eco_lev01/
  us_eco_lev02/
  koppen_cleaned/
```

4.2.7 Step 0.4.1 — Update project structure refactor report

- **Action:** Update project structure refactor report (this Rmd). Tasks: Reorganize report sections. Update file folder structure metadata. Create missing file folders in FFA.
- **Reason/Before:** Unclear project workflow and documentation. Differences between project file structure and file structure metadata.
- **After:** Improves readability, clarity, consistency of report. Provides consistent naming across project

4.2.8 Step 0.4.2 — Identify and document covariates

- **Action:** Identify covariates. Create and validate covariate metadata,
- **Reason/Before:** Covariates not structured spatially or by domain.
- **After:** The covariate framework is hierarchically organized across five spatial scales, ranging from station-level (scale 0) to local, subregional, regional, and macroregional levels (scales 1–4), and four major domains: Climate, Land Cover, Topography, and Watershed Metrics. The covariate metadata documents all variables used in regional skew modeling and spatial stratification.

Table 2: Table: Covariate Count Organized by Domain and Scale

Level	Scale	Extent	Climate	Land Cover	Topography	Watershed	Total
0	Station-level	Point-derived	2	0	1	1	4
1	Macroregional	Custom Macrozone	3	4	3	0	10
2	Regional	Level II Ecoregion	4	4	4	5	17
3	Subregional	Level III Ecoregion	6	2	4	3	15
4	Local	NHD+ catchments	3	2	8	4	17

4.2.9 Step 0.4.3 — Validate and document Notes folder

- **Action:** Added notes folder to Git
- **Before/Reason:** Notes folder was not included in Git, as clear notes on method had not been compiled
- **After:** Notes folder contains discussion how zonal statistics will be calculated and other notes regarding covariates.

4.2.10 Step 0.4.4 — Document raw data/ folder structure / naming strategy

- **Action:** Documented a naming strategy in reports/milestones_01_download scripts
- **Before/Reason:** Inconsistent naming strategy – led to a lot of confusion
- **After:** Adopted a Subdirectory Strategy for folders and a source[layer]/[year|version]_[status].[ext] strategy for naming.

4.2.11 Step 0.3.7 – Refactor and Modularize

- **Action:** Created sub-issue v0.4 under Milestone 00. Moved script-related steps (e.g., downloading covariate data) to **Milestone 01 – Data Download and Preparation**. Closed the sub-issue after restructuring.
- **Reason (Before):** Covariate identification and metadata creation required their own distinct milestone. The original milestone was too broad.
- **Result (After):** Project structure is now better modularized by task and milestone. Documentation and code are easier to maintain.

4.2.12 Step 0.3.8 – Update GitHub Milestone

- **Action:** Updated GitHub Milestone 00 title and description to reflect changes in scope and tagging.

4.2.13 Step 0.3.9 – Final Edits to Documentation

- **Action:** Updated this .Rmd file with a GitHub-friendly YAML header. Ensured clarity, TOC, and version parameter (params\$version: “v0.4”).

4.2.14 Step 0.3.10 – README Update

- **Action:** Edited README.Rmd and README.md to reflect the current project structure and meta-data paths.

4.2.15 Step 0.3.11 – Git Commit and Tag

- **Action:** Committed changes to Git, including output and documentation files. Tagged the repository as v0.4.